A Confirmatory Study of Complementary-Filtered Residual World Models under Synthetic Context Corruption

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Abstract

Long-horizon world-model prediction can fail before rollout begins: a nominal physical prior may be biased under parameter shift, whereas recent measurements may be noisy or missing. We introduce the Complementary-Filtered Residual World Model (CFR-WM), which applies a prespecified gain-0.5, mask-exact complementary update over the observed context and initializes a probabilistic physics-residual transition for 25-step open-loop forecasting. The contribution is this auditable composition and its confirmatory evaluation, not a claim that complementary or Kalman-style filtering is new or universally optimal. We train 440 models spanning five controlled nonlinear systems, three clean/corrupted/parameter-shift regimes, eight independent seed-level realizations, five broad comparators, and five gain/training-horizon sensitivities. The frozen primary estimand is the mean of seed-specific relative NRMSE-25 changes against a preselected physics-guided neural network style comparator (PGNN-style), with all 15 system–regime cells equally weighted. The estimated relative improvement was 8.97% (95% bootstrap CI [6.88%, 10.95%]); the absolute proposed-minus-PGNN-style difference was -0.0098 (95% CI [-0.0121, -0.0075]), and the exact one-sided sign-flip value for a 5% practical margin was $\boldsymbol{p = 0.00391}$. Thus H1 was supported, but the prespecified H2 robustness claim was not supported because clean noninferiority failed: corrupt/shifted cells improved by 12.05%, whereas clean cells changed by -13.08%. The evidence is limited to transition prediction on a finite synthetic-mechanistic benchmark with known future controls; it does not establish policy quality, visual world modeling, or real-robot transfer.

Keywords: world models, physics-guided machine learning, complementary filtering, long-horizon prediction, missing observations, distribution shift

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1 Introduction

World models compress observations and actions into a state from which future outcomes can be predicted. Their appeal is clearest when direct interaction is costly: imagined trajectories can support planning, control, or data-efficient reinforcement learning. DreamerV3, TD-MPC2, IRIS, and STORM illustrate the range of recurrent, implicit, and transformer dynamics used in modern agents [1–4]. However, benchmark return does not by itself establish that a transition model has a reliable forecast origin when recent observations are noisy or missing. Reviews of model-based reinforcement learning and real-world deployment treat predictive quality, partial observability, distribution shift, and safety as distinct problems [5–7].

Known dynamics offer useful inductive bias but do not eliminate the forecast- origin problem. A nominal simulator can be interpretable and data-efficient, yet parameter drift, unmodeled friction, discretization, and actuator error make its state estimate biased. A direct measurement is current but may be corrupted or absent. Physics-guided learning combines mechanistic and neural components through losses, features, hybrid solvers, and probabilistic state models [8–11]. KalmanNet and data-assimilation research further show that learned or physics-aware filtering is established prior art [12, 13]. The narrower unresolved question is whether a deliberately simple, prespecified complementary update can provide a more robust starting state for a long-horizon physics-residual world model under corrupted context.

That question separates three operations that are often evaluated only as a single end-to-end score. A representation maps the available history into a state, a context update decides how strongly to trust a new observation, and a transition propagates the resulting forecast origin. Improving the transition cannot repair an arbitrarily biased origin, while an accurate context update cannot prevent a misspecified transition from accumulating error. Missingness makes the distinction sharper: last-value imputation supplies a numerical input but does not make that value contemporaneous, whereas nominal propagation remains current but inherits parameter error. Learned filters can adapt this trade-off, but their extra flexibility also creates a mechanism-identification question: an apparent gain over a deterministic physics-guided model may come from the residual transition rather than from the learned update itself. We therefore treat equal-trust filtering as a falsifiable engineering choice and compare it with both no-update and measurement-dominant sensitivities.

We study this question with the Complementary-Filtered Residual World Model (CFR-WM). Within the observed eight-step context, nominal physics predicts the next state and a fixed gain of 0.5 combines that prediction with each available measurement; missing channels receive exactly zero measurement gain. The final filtered

state initializes a probabilistic nominal-physics plus bounded neural- residual transition. This transition is trained by multi-step Gaussian likelihood and rolled out freely for 25 steps. The fixed update avoids an unsupported claim that a learned gain is necessary, while gain 0, 0.25, 0.75, and 1 sensitivities test whether the result is specific to equal trust.

A second gap is evidential rather than architectural. Flexible synthetic benchmarks make it inexpensive to try many candidate mechanisms, but the same flexibility can silently turn development outcomes into confirmation. We use a versioned firewall: two rejected candidates and the normalization defect that invalidated the first analysis remain archived; a new eight-seed grid was frozen only after candidate selection; and the runner refuses changes to code, tuning, methods, seeds, metrics, or decision thresholds. The primary endpoint tests a practical 5% margin directly rather than declaring success from an interval that merely excludes zero. This design makes a negative result as interpretable as a positive one and prevents favorable system-level cells from rescuing a failed aggregate hypothesis.

The study is scoped to transition prediction. Five controlled nonlinear systems—a damped oscillator, Duffing oscillator, Van der Pol oscillator, torque-driven pendulum, and cart–pole—are evaluated under clean ID, corrupted ID, and corrupted parameter-shift regimes. The analytical benchmark makes trajectories, corruption, and shift auditable, but it does not establish image understanding, policy quality, hardware safety, or real-robot transfer.

Our contributions are:

1. a mask-exact complementary context update whose fixed gain and missing- channel behavior are frozen before training, coupled to a probabilistic physics-residual free rollout;
2. an exact error decomposition that states when complementary updating can reduce forecast-origin error, without claiming universal filter optimality;
3. an approximately parameter-matched evaluation against five broad transition baselines, plus five prespecified gain and training-horizon sensitivities over 15 fixed benchmark cells; and
4. a fail-closed confirmatory design with eight independent seed clusters, paired repeated-measurement resampling, direct testing of a 5% relative- improvement margin, attainable multiplicity control, and archived negative development decisions.

The primary hypothesis was frozen before any confirmatory seed was generated: the mean seed-specific NRMSE-25 improvement over a preselected PGNN-style comparator must be at least 5%. Support requires both a bootstrap lower bound of 5% and a one-sided exact seed sign-flip test against that same margin. This tests the claimed practical magnitude rather than combining a 5% point estimate with a weaker test against zero.

The remainder of the paper positions the method against recent world-model, physics-guided, and state-estimation work; defines the causal filter and probabilistic free rollout; specifies the fixed benchmark and inferential contract; and reports aggregate, regime, comparator, and gain-sensitivity results. The discussion then

separates measured evidence from conditional mathematical properties and identifies the additional visual, control, and hardware experiments required for broader embodied-world-model claims.

2 Related work

2.1 World models and transition backbones

Contemporary world models span recurrent stochastic state-space models, discrete autoregressive transformers, implicit latent dynamics, and newer sequence architectures. DreamerV3 couples recurrent latent dynamics with normalization and imagined behavior learning, while TD-MPC2 uses decoder-free latent prediction and local trajectory optimization [1, 2]. IRIS and STORM demonstrate transformer-based world models in the Atari-100k protocol [3, 4]. UniZero, HarmonyDream, and curious replay explore generalized planning, multitask harmonization, and adaptation through replay [14–16]. Scaling and regularization remain active design dimensions [17].

Applications in driving and manipulation add realistic observation and planning demands: Think2Drive and Dream-to-Drive use latent predictive models in CARLA-like settings, DINO-WM plans over pretrained visual features, and recent reviews organize learned dynamics for robotic manipulation [18–21]. QT-TDM combines transformer dynamics with autoregressive value prediction [22]. GLAM separates global and local variation using Mamba-style sequence modules [23]. These systems establish important transition families, but their published task returns, image-token losses, or planning scores are not protocol-equivalent to calibrated continuous-state forecasts. We therefore implement compact, approximately parameter-matched transition adaptations of recurrent stochastic, SimNorm, and transformer families, rather than claiming direct reproduction of their complete agents.

The rapidly moving literature also sharpens the novelty boundary. The 2025 Prismatic World Model uses a mixture of experts to represent compositional hybrid modes, and the 2026 HaM-World preprint combines a soft Hamiltonian subspace, residual dynamics, and selective memory [24, 25]. Another 2026 preprint argues that structural assumptions can certify a useful prediction horizon [26]. These non-archival works are treated as recency stress tests, not established baselines. Their mode assignment, Hamiltonian structure, or equivariant certificates differ from using a fixed mask-exact complementary update to initialize a probabilistic residual rollout. Broader surveys likewise show that "world model" covers representation, dynamics, planning, and agent-learning questions that must not be conflated [27].

2.2 Physics-guided dynamical learning

Physics-guided learning can constrain outputs, embed equations, combine solvers with neural corrections, or assimilate uncertain observations. Reviews emphasize that no single integration pattern dominates across scientific problems [10, 28, 29]. Differentiable modeling exposes mechanistic components to gradient-based learning, while causal or horizon-aware optimization is important when temporal residuals compound [11, 30, 31].

Several works are especially close to our transition setting. Physics-guided neural networks have been evaluated on nonlinear dynamics by injecting partial physics into a learned architecture [32]. Physics-guided Deep Markov Models integrate state-space physics with probabilistic latent inference [33]. KalmanNet learns Kalman-style state updates when dynamics are only partially known, establishing that learned filtering is itself not a novelty claim [12]. Hybrid online models have combined physical and data-driven components for electric vehicles, while related dynamic-state estimators couple both sources in integrated energy systems [34, 35]. Machine learning with data assimilation further connects dynamical estimation and uncertainty quantification [13]. Domain studies, such as physics-guided lake-temperature simulation, demonstrate the usefulness of scientific priors but also underline that conclusions remain system-specific [36].

Our method is not novel merely because it contains physics, a residual, a variance, or a filter. The tested composition is narrower: a gain frozen independently of confirmation performs mask-exact context correction, and the resulting forecast origin initializes a multi-step probabilistic physics-residual world model. The gain sensitivities and the failed learned-gain development gate keep this claim separate from filter-optimality or adaptive-gain claims.

2.3 Predictive uncertainty and calibration

Deep-learning uncertainty can represent data noise, model uncertainty, or their approximation, and a low negative log likelihood need not imply calibrated intervals [37]. Sequence-model studies in rainfall–runoff prediction provide useful precedents for evaluating coverage and interval width together, rather than reporting one score [38]. Bayesian calibration work explicitly distinguishes model-form uncertainty in dynamical systems [39], while automated calibration has been studied for wave downscaling [40]. Conformal prediction offers a distribution-light route to uncertainty sets for dynamic biological systems [41]; it is complementary to, not implemented by, the present Gaussian moment-matching model. We consequently report Gaussian NLL, empirical 90% coverage, mean interval width, and calibration error over nine central-coverage levels.

3 Complementary-filtered residual world model

3.1 Forecasting problem

Let $\mathbf{x}_t \in \mathbb{R}^{d_x}$ be a physical state and $\mathbf{u}_t \in \mathbb{R}^{d_u}$ a recorded control. The observation available to the model is

$$\mathbf{y}_t = \mathbf{m}_t \odot (\mathbf{x}_t + \boldsymbol{\epsilon}_t), \tag{1}$$

where $\mathbf{m}_t \in \{0,1\}^{d_x}$ is a channel-wise availability mask and $\boldsymbol{\epsilon}_t$ is observation noise. Missing entries are filled causally by the latest available value, producing $\tilde{\mathbf{y}}_t$; the mask and elapsed time since observation $\boldsymbol{\delta}_t$ remain explicit inputs, so an imputed value is never represented as a fresh measurement.

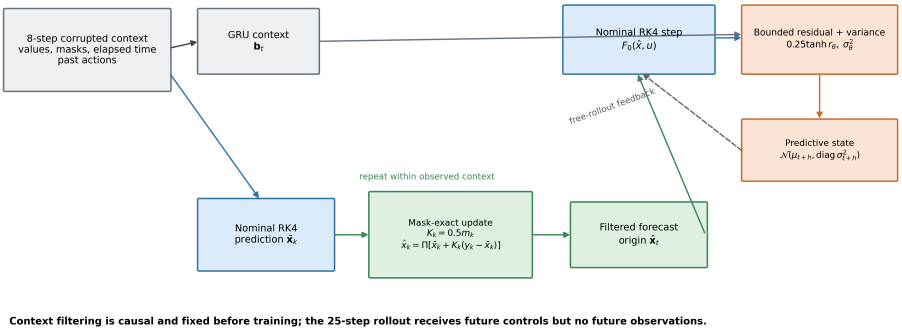


Fig. 1 Causal data flow of CFR-WM. A fixed complementary update is applied only inside the observed context. Missing channels receive exactly zero measurement gain. The filtered forecast origin then initializes a probabilistic nominal-physics plus neural-residual transition, which is rolled out freely for 25 steps using known future controls and no future observations.

At forecast origin t, the model receives eight context states and their past controls,

$$\mathcal{C}_t = \{\tilde{\mathbf{y}}_{t-7:t}, \mathbf{m}_{t-7:t}, \boldsymbol{\delta}_{t-7:t}, \mathbf{u}_{t-7:t-1}\}, \tag{2}$$

together with known future controls $\mathbf{u}_{t:t+H-1}$. It estimates a diagonal predictive distribution for $\mathbf{x}_{t+1:t+H}$ under free rollout, with $H = 25$. Future states or observations are never fed back. All learned operations use affine statistics estimated from the training split; the nominal equations are evaluated in physical coordinates and then returned to normalized coordinates.

3.2 Mask-exact complementary context filter

Let F_0 be one fourth-order Runge–Kutta step of a nominal controlled differential equation. Starting from the first imputed context vector, the filter alternates nominal prediction and measurement update. For context index $k > t - 7$,

$$\bar{\mathbf{x}}_k = F_0(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_{k-1}), \tag{3}$$

$$\mathbf{K}_k = \alpha \mathbf{m}_k, \qquad \alpha = 0.5, \tag{4}$$

$$\hat{\mathbf{x}}_k = \Pi[\bar{\mathbf{x}}_k + \mathbf{K}_k \odot (\tilde{\mathbf{y}}_k - \bar{\mathbf{x}}_k)]\,. \tag{5}$$

Thus, before projection, an observed channel receives equal weight from the nominal prediction and the current measurement, while a missing channel has exactly zero measurement correction. The projection Π is the identity except for the sine–cosine pendulum representation, which is returned to the unit circle after each update. That joint projection can alter either pendulum channel even when its own measurement gain is zero, so the implementation does not claim channel-wise equality to nominal propagation after projection. The gain is fixed before training and has no access to validation or test outcomes.

In parallel, a GRU encodes $[\tilde{\mathbf{y}}_k, \mathbf{m}_k, \log(1+\boldsymbol{\delta}_k), \mathbf{u}_{k-1}]$. Its last hidden state supplies trajectory context to the forecast transition; it does not alter the fixed measurement gain. The implementation retains the same unused gain-head parameterization in every fixed-gain sensitivity model, so comparisons at gains 0, 0.25, 0.5, 0.75, and 1 do not change nominal capacity. We do not claim that complementary or Kalman-style filtering is new; KalmanNet and modern data-assimilation work establish the broader prior art [12, 13]. The tested contribution is the use of a prespecified, mask-exact context update as the forecast origin of a physics-residual probabilistic world model.

3.3 Probabilistic physics-residual rollout

The last filtered context state initializes a free rollout. At step h,

$$\boldsymbol{\mu}^{p}_{t+h} = F_0(\hat{\mathbf{x}}_{t+h-1}, \mathbf{u}_{t+h-1}), \tag{6}$$

$$[\mathbf{r}_{t+h}, \ell_{t+h}] = R_\theta([\hat{\mathbf{x}}_{t+h-1}, \mathbf{u}_{t+h-1}, \mathbf{b}_{t+h-1}]), \tag{7}$$

$$\boldsymbol{\mu}_{t+h} = \Pi\big[\boldsymbol{\mu}^{p}_{t+h} + 0.25\tanh(\mathbf{r}_{t+h})\big], \tag{8}$$

$$\boldsymbol{\sigma}^2_{t+h} = \max\big\{\exp(\mathrm{clip}(\ell_{t+h}, -8, 4)), 10^{-6}\big\}. \tag{9}$$

The fixed residual scale bounds a single normalized correction but does not guarantee a bounded trajectory because errors can accumulate. The predicted mean becomes the next state, and a GRUCell updates $\mathbf{b}$ from that mean and the known control. No sampled state or future target enters the recurrence.

The training loss is the mean diagonal Gaussian negative log likelihood over all H steps and state channels,

$$\mathcal{L}_{\mathrm{NLL}} = \frac{1}{2Hd_x}\sum_{h=1}^{H}\sum_{q=1}^{d_x}\left[\log(2\pi\sigma^2_{hq}) + \frac{(x_{hq}-\mu_{hq})^2}{\sigma^2_{hq}}\right]. \tag{10}$$

The one-step ablation applies the same loss only at $h = 1$ but is still evaluated by a 25-step free rollout.

3.4 Causal execution and computational cost

For each window, execution follows four deterministic stages. First, the GRU encodes the imputed context together with masks, elapsed times, and past controls. Second, Equation (5) advances the nominal state through the same context and applies the mask-exact correction after each available observation. Third, the final hidden state and filtered state jointly initialize the residual recurrence. Fourth, Equation (9) is iterated for every known future control; predicted means, rather than samples or targets, feed the next step. Training backpropagates the multi-step NLL through this entire causal graph. Evaluation uses the identical graph with gradients disabled, so teacher forcing is absent from both validation and test rollouts.

Let T_c denote context length, H forecast horizon, d_b recurrent width, and C_{F_0} the cost of one nominal RK4 step. Dense recurrent operations cost $O((T_c + H)d_b^2)$ per trajectory, while nominal propagation costs $O((T_c + H)C_{F_0})$. Diagonal mean and

variance storage is $O(Hd_x)$; recurrent working memory at inference is $O(d_b + d_x)$. Thus inference grows linearly in context and horizon. The analysis does not claim asymptotic superiority over the comparators: the fixed filter adds nominal context steps, and measured latency is reported as an implementation property rather than a theoretical speed advantage.

3.5 Error decomposition and claim boundary

For an observed scalar channel on which Π is the identity, write the nominal pre-update error as $e_p = \bar{x} - x$ and measurement error as $\epsilon = y - x$. Before any non-identity projection, Equation (5) gives the exact post-update error

$$e_f = (1 - \alpha)e_p + \alpha\epsilon. \tag{11}$$

If these errors are zero-mean and uncorrelated, then

$$\mathbb{E}[e_f^2] = (1 - \alpha)^2\mathbb{E}[e_p^2] + \alpha^2\mathbb{E}[\epsilon^2]. \tag{12}$$

Equal trust is therefore not universally optimal: it helps only when the two error sources are sufficiently balanced, and a missing channel receives no direct measurement-error term because its gain is zero. For the pendulum's joint sine–cosine projection, an additional coupled projection error must be accounted for, so Equation (12) is not asserted channel-wise after projection. This conditional identity, rather than an optimal-filter claim, motivates the fixed-gain sensitivity analysis.

If the rollout transition is locally L-Lipschitz and incurs local error η_h, the standard recursion yields

$$\|\hat{\mathbf{x}}_{t+H} - \mathbf{x}_{t+H}\| \le L^H\|\hat{\mathbf{x}}_t - \mathbf{x}_t\| + \sum_{h=0}^{H-1} L^{H-1-h}\eta_h. \tag{13}$$

The context filter can reduce only the initial-error term; the residual model must control subsequent local errors. Neither relation provides an unconditional stability guarantee or supports claims about visual representation learning, policy quality, or real-robot transfer.

4 Experimental design

4.1 Controlled systems, corruption, and shift

We evaluate five action-conditioned nonlinear systems. The damped oscillator has state $(q, \dot{q})$ and dynamics $m\ddot{q} + c\dot{q} + kq = u$; the Duffing oscillator follows $\ddot{q} + c\dot{q} + \alpha q + \beta q^3 = u$; and the controlled Van der Pol system follows $\ddot{q} = \mu(1 - q^2)\dot{q} - q + g_u u$. The pendulum is represented by $(\sin\theta, \cos\theta, \dot{\theta})$, and the cart–pole by $(x, \dot{x}, \theta, \dot{\theta})$. Reference trajectories use fourth-order Runge–Kutta integration with four internal substeps. Initial conditions, bounded filtered stochastic controls, and physical parameters are

generated from recorded seeds. Shifted parameter intervals are prespecified and disjoint from training intervals; unit tests verify that separation.

Each seed–system dataset contains 256 training, 64 validation, and 96 ID test trajectories, plus 96 independently generated OOD test trajectories. A trajectory has 72 states. Windows contain eight context states and 25 forecast transitions; training and validation strides are eight and 16, respectively, whereas evaluation uses one window per trajectory. Split identifiers encode system, parameter regime, split, trajectory index, and generation seed. No trajectory crosses splits.

The three deployment regimes are (i) clean ID context, (ii) ID context with feature-wise Gaussian noise at 8% of the mean training-state standard deviation and a nominal 25% independent missing-at-random probability, and (iii) the same corruption with OOD physical parameters. Training context uses 3% noise and a nominal 10% missing probability. The first time point of every generated trajectory is forced observed in every channel so causal last-value imputation is defined; masks at all subsequent trajectory time points are sampled independently. A later sliding window can therefore begin at an unobserved, causally imputed context value. Masks and channel-wise elapsed times are retained. ID-clean and ID-corrupt evaluations reuse the same underlying 96 trajectories, permitting repeated-measurement resampling. OOD identifiers are required to be disjoint from ID identifiers.

4.2 Comparators, sensitivities, and development firewall

The five broad comparators cover recurrent, normalized, transformer, and physics-guided transition families. **RSSM-style** is a compact recurrent stochastic-style transition. **SimNorm-style** applies grouped simplex normalization to its recurrent representation, adapting a representation principle used by scalable continuous-control world models [2]. **Transformer-style** uses a two-layer causal context transformer followed by recurrent free rollout [3]. **PGNN-style** adds a deterministic neural correction to a nominal physics step [32]. **PG-DMM-style** is a probabilistic physics-residual transition inspired by physics-guided Deep Markov modeling [33]. These are protocol-matched transition implementations, not reproductions of published end-to-end agents or their reported returns.

The proposed **CFR-WM** uses gain 0.5. Four fixed-gain sensitivities use $\alpha \in \{0, 0.25, 0.75, 1\}$, and **CFR-one-step** changes only the training horizon. All six CFR variants have hidden width 72 and identical parameterization; even the inactive fixed-gain head is retained. Comparator widths are 80 for RSSM-style and SimNorm-style, 48 for Transformer-style, and 84 for PGNN-style and PG-DMM-style.

Candidate selection was separated from confirmation. A first development model, reliability-gated RGP-WM, failed its corrected primary hypothesis after an audit replaced a trajectory-local NRMSE denominator by a training-only scale. A structurally distinct learned-gain PFR-WM then improved over retrained the PGNN-style comparator on development seed 11, but exceeded its prespecified gain-0.5 matched ablation by only 0.223%, below the frozen 2% mechanism threshold. Learned gain was therefore rejected and is absent from every confirmatory seed. The gain-0.5

comparator from that failed mechanism test became the versioned CFR-WM candidate before confirmation. Original development outputs, corrected re-evaluations, and failure decisions remain archived; none is pooled with the confirmatory analysis.

4.3 Optimization and endpoint definitions

All models use AdamW, batch size 128, gradient clipping at norm 5, at most 60 epochs, and early stopping after ten epochs without a validation-NLL improvement of 10^{-5}. The pre-confirmation tuning archive contains the same six-trial validation budget for every baseline–system pair. Each baseline uses its own frozen winner. CFR-WM and all CFR sensitivities inherit the frozen PGNN-style winner for the same system, preventing an additional candidate-specific search. No confirmatory result can alter a hyperparameter.

The primary endpoint is NRMSE through horizon 25. Per-feature RMSE is divided by that feature's standard deviation estimated from the training split only, then averaged over features. Because stored targets and predictions are already in those normalized coordinates, the evaluation divisor is one. This avoids estimating a denominator from the test trajectory and remains defined at horizon one. Secondary accuracy endpoints are cumulative NRMSE at horizons 1, 5, and 10 and MAE through horizon 25. Probabilistic endpoints are diagonal Gaussian NLL, empirical 90% coverage, mean 90% interval width, and mean absolute central-interval calibration error over nominal levels 0.1–0.9. A trajectory is catastrophic when any feature-step absolute error exceeds three training-set standard deviations.

4.4 Frozen confirmatory inference

The confirmation grid contains 11 methods, eight untouched seeds $(41, 53, 67, 79, 83, 97, 109, 113)$, five systems, and three regimes: 440 training runs, 1,320 aggregate cells, and 126,720 paired trajectory rows. One deterministic dataset fingerprint is shared by all methods within each seed–system group, and all 40 such fingerprints must be distinct. The eight seed-level training/data realizations are the top-level independent clusters; trajectories are nested observations. Systems and regimes are treated as a fixed benchmark, so inference does not imply sampling from a population of all physical systems.

For a seed s, let B_s and C_s be the PGNN-style and CFR-WM NRMSE-25 means after equal weighting of all 15 system–regime cells. The primary estimand is

$$\Delta_{\mathrm{rel}} = \frac{1}{8}\sum_{s=1}^{8} 100\frac{B_s - C_s}{B_s}. \tag{14}$$

A 5,000-replicate bootstrap resamples seeds globally with replacement. Within each selected seed and system it resamples paired trajectory identifiers; ID-clean and ID-corrupt reuse the same sampled identifier vector, while the disjoint OOD set is resampled independently. Every system–regime cell is retained, and B_s, C_s, and their ratio are recomputed in each replicate. The interval is the two-sided percentile 95% interval.

H1 states that CFR-WM improves on the preselected PGNN-style comparator by at least 5%. Support requires finite metrics, a lower relative-improvement interval endpoint of at least 5%, and a one-sided exact sign-flip $p < 0.05$ for the seed effects after subtracting the 5% margin. All 2^8 sign configurations are enumerated under a prespecified symmetry assumption. Confirmation success is determined by H1 alone.

H2 is tested only if H1 succeeds. It requires at least 5% improvement when pooling ID-corrupt and OOD-corrupt fixed cells, and clean-regime noninferiority with signed relative-improvement margin -5%. Both components must meet their respective bootstrap bound and one-sided margin test, with all metrics finite. The five broad CFR-versus-comparator contrasts form one Holm family; the five CFR sensitivity/ablation contrasts form another. With eight clusters, the minimum two-sided exact raw and five-test Holm-adjusted values are $2/2^8 = 0.0078125$ and 0.0390625. The 15 system–regime contrasts report paired effects and unadjusted bootstrap intervals only; they are explicitly descriptive and carry no cell-level significance claim.

4.5 Fail-closed reproducibility controls

The machine-readable contract was frozen before the first confirmatory dataset. It pins method and seed grids, data counts, optimization, estimands, margins, multiplicity families, output location, and SHA-256 fingerprints for training code, analysis code, and the tuning archive. The dedicated runner accepts no overrides. Before data generation it checks all contract fields and hashes; partial resume is allowed only for an exact provenance-bound subset with paired JSON and checkpoint artifacts. The generic experiment CLI rejects every frozen seed and the reserved output directory before loading tuning or generating data. The analyzer independently revalidates the contract, current code, payload configurations, 40 distinct dataset fingerprints, exact row grid, paired trajectory identifiers, and finite values. Intermediate confirmatory metrics are not inspected.

The first invocation of the frozen analyzer stopped before any bootstrap, decision, or result file because its generic loader omitted the already stored top-level `parameter_count` field required only by the descriptive model-size summary. Before inspecting any performance output, we froze a versioned analysis amendment that performs one validated many-to-one join on method, seed, and system, then calls the unchanged confirmatory and statistical functions. It also rechecks all 440 JSON/checkpoint pairs and refuses to overwrite an analysis directory. The original contract, failed zero-output logs, amendment, and their hashes are retained; no outcome, estimand, margin, random seed, bootstrap count, or decision rule changed.

Execution uses Python 3.11.15 and PyTorch 2.11.0+cu130 on an NVIDIA RTX 5070 Laptop GPU with 8,151 MiB reported memory and an AMD Ryzen 9 9955HX processor. The sequential runner stops at 80% of an eight-hour budget, writes each condition atomically, and never synthesizes missing performance values.

Table 1 Confirmatory accuracy over the fixed five-system, three-regime benchmark. Entries are means ± SD across eight seed-level, cell-balanced aggregates. Bold marks the lowest aggregate NRMSE-25; it does not by itself denote confirmatory support. Parameter counts are means over the five system-specific input dimensions; all methods received the same six-trial validation search budget.

Method	Mean parameters	NRMSE-1	NRMSE-10	NRMSE-25
CFR-WM	57694	0.039 ± 0.001	0.057 ± 0.002	**0.100 ± 0.004**
CFR-gain-0.25	57694	0.037 ± 0.001	0.059 ± 0.002	0.103 ± 0.004
CFR-gain-0.75	57694	0.047 ± 0.001	0.065 ± 0.002	0.107 ± 0.005
PGNN-style	61275	0.060 ± 0.001	0.070 ± 0.002	0.110 ± 0.003
CFR-gain-1	57694	0.059 ± 0.001	0.073 ± 0.002	0.114 ± 0.004
PG-DMM-style	61493	0.063 ± 0.001	0.075 ± 0.002	0.114 ± 0.003
RSSM-style	62485	0.073 ± 0.002	0.090 ± 0.005	0.123 ± 0.006
Transformer-style	59708	0.074 ± 0.002	0.094 ± 0.004	0.132 ± 0.009
SimNorm-style	62485	0.074 ± 0.002	0.098 ± 0.006	0.138 ± 0.013
CFR-gain-0	57694	0.081 ± 0.003	0.100 ± 0.003	0.144 ± 0.005
CFR-one-step	57694	0.040 ± 0.001	0.136 ± 0.050	0.278 ± 0.104

5 Results

5.1 Integrity checks and aggregate accuracy

The confirmatory analyzer accepted the complete grid of 440 trained models and 126,720 trajectory-level rows. All prespecified metrics were finite (yes), and the 40 seed–system groups produced 40 distinct data fingerprints while sharing one fingerprint across methods within each group. These checks establish grid completeness and pairing; they are not performance evidence.

Table 1 reports the seed-balanced mean and standard deviation for horizons 1, 10, and 25. Across all 11 methods, CFR-WM had the lowest aggregate NRMSE-25 (0.0998). CFR-WM's mean was 0.0998, compared with 0.1096 for the preselected PGNN-style comparator. This ordering is descriptive: confirmatory support depends on paired seed-level effects and frozen practical margins, not on the lowest pooled mean alone. The confirmatory estimand does not follow solely from these pooled means: it averages eight within-seed relative effects after giving every fixed system–regime cell equal weight.

Frozen primary estimand. For each seed, PGNN-style and CFR-WM NRMSE-25 are first averaged with equal weight over the 15 prespecified system–regime cells; their relative improvement is then computed. The confirmatory point estimate is the arithmetic mean of those eight seed-level relative improvements, not the ratio of the two pooled means in Table 1.

5.2 Primary and gated robustness hypotheses

Against PGNN-style, the seed-mean relative NRMSE-25 improvement was 8.97% (95% paired fixed-cell bootstrap CI [6.88%, 10.95%]). In absolute units, CFR-WM minus PGNN-style was -0.0098 (95% CI [-0.0121, -0.0075]). The exact one-sided sign-flip test

Table 2 Seed-level primary effects after equal weighting of the 15 frozen benchmark cells. Positive values favor CFR-WM.

Seed	CFR-WM improvement over PGNN-style, %
41	+5.29
53	+12.76
67	+12.01
79	+10.49
83	+9.69
97	+6.66
109	+9.32
113	+5.56

against the same 5% practical margin gave $p = 0.00391$. Under the frozen conjunction of a 5% interval lower bound, $p < 0.05$, and finite metrics, H1 was supported. This decision is not changed by rankings against other comparators or by favorable individual cells.

For the H2 components, the combined corrupt-ID and corrupt-OOD regimes had a relative effect of 12.05% (95% CI [10.51%, 13.49%], one-sided 5%-margin $p = 0.00391$). ID-clean had an effect of -13.08% (95% CI [-20.60%, -5.67%], one-sided signed -5%-margin $p = 0.95703$). Because H2 is hierarchically gated by H1 and requires both component rules, the robustness hypothesis was not supported.

Table 2 exposes the eight independent cluster effects rather than only their mean. Every seed exceeded the 5% practical threshold, although the exact test still treats the seed effects as the inferential units.

5.3 Comparator and sensitivity evidence

Table 3 reports every prespecified broad comparison. Positive relative effects favor CFR-WM. The confidence intervals retain all fixed cells, whereas the exact two-sided seed sign-flip values are Holm-adjusted as one five-test family. These comparisons contextualize H1 but do not replace its preselected PGNN-style contrast or its 5% margin.

The PGNN-style row repeats the primary point estimate but uses the frozen comparator-family bootstrap seed (20260808), whereas H1 uses its prespecified primary seed (20260805). The estimand and 5,000-replicate resampling scheme are otherwise identical; the separately sampled interval endpoints differ by at most 0.01 percentage point after rounding, and only the primary interval enters the H1 decision.

The lowest aggregate NRMSE-25 among gains 0, 0.25, 0.5, 0.75, and 1 occurred at gain 0.50. Relative to the one-step-trained variant, CFR-WM's effect was 59.38% (95% CI [48.88%, 68.88%], Holm $p = 0.03906$). Table 4 gives the complete second five-test family; therefore the gain sweep is reported even when it does not favor the proposed value.

Secondary probabilistic metrics are shown in Table 5. They are diagnostics rather than alternate confirmation endpoints: lower Gaussian NLL, calibration error, or

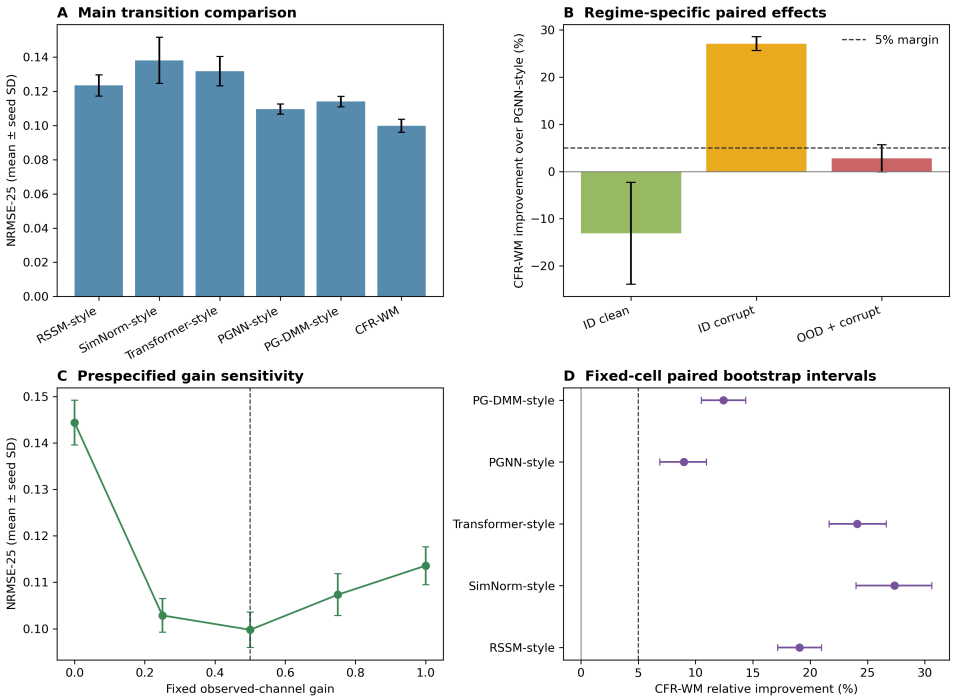


Fig. 2 Locked confirmatory summary. (A) Aggregate transition accuracy for the six main methods. (B) Paired CFR-WM effects against PGNN-style by regime; positive values favor CFR-WM. (C) Prespecified fixed-gain sensitivity. (D) Paired fixed-cell intervals against each broad comparator. Error bars in A–C are seed-level SD; D uses the nested 95% bootstrap interval.

Table 3 CFR-WM relative NRMSE-25 effects against broad comparators. Exact p values test a zero seed-level difference; Holm adjustment is confined to this prespecified family of five.

Reference	Relative effect [95% CI], %	Exact p	Holm p
RSSM-style	+19.08 [+17.16, +20.98]	0.0078	0.0391
SimNorm-style	+27.34 [+24.00, +30.62]	0.0078	0.0391
Transformer-style	+24.09 [+21.65, +26.64]	0.0078	0.0391
PGNN-style	+8.97 [+6.89, +10.96]	0.0078	0.0391
PG-DMM-style	+12.44 [+10.51, +14.39]	0.0078	0.0391

catastrophic-rollout rate cannot rescue a failed H1, and diagonal-Gaussian calibration does not imply a correct joint forecast distribution.

5.4 Fixed-cell heterogeneity

Figure 3 retains all 15 system–regime effects. By the frozen descriptive guard, CFR-WM was more than 10% worse than PGNN-style in 3 cells. All three were ID-clean: cart–pole (+47.3% relative change), controlled Van der Pol (+19.4%), and Duffing (+24.4%). Conversely, 5 of 5 ID-corrupt cells favored CFR-WM, with descriptive

Table 4 CFR-WM relative NRMSE-25 effects against gain and training-horizon sensitivities. Positive values favor the proposed gain-0.5 multi-step model.

Reference	Relative effect [95% CI], %	Exact p	Holm p
Gain 0	+30.89 [+28.95, +32.67]	0.0078	0.0391
Gain 0.25	+2.97 [+0.88, +5.27]	0.0312	0.0391
Gain 0.75	+6.97 [+4.77, +9.42]	0.0078	0.0391
Gain 1	+12.13 [+10.31, +13.92]	0.0078	0.0391
One-step training	+59.38 [+48.88, +68.88]	0.0078	0.0391

Table 5 Secondary aggregate diagnostics for the six broad methods. Rates and scores are seed-balanced over all fixed cells and are not separately tested.

Method	Parameters	Gaussian NLL	Calibration error	Catastrophic rate
RSSM-style	62485	-0.183	0.214	0.01%
SimNorm-style	62485	-0.246	0.206	0.01%
Transformer-style	59708	-0.279	0.202	0.01%
PGNN-style	61275	-0.385	0.248	0.00%
PG-DMM-style	61493	0.029	0.245	0.00%
CFR-WM	57694	0.013	0.228	0.00%

changes from -13.3% to -34.8%. This pattern explains why an aggregate H1 success coexists with failure of clean-regime noninferiority. No cell-level p value is reported: with eight seed clusters, those contrasts were not assigned confirmatory multiplicity claims. The exact estimates and unadjusted bootstrap intervals are archived in the appendix table and machine-readable CSV.

6 Discussion

The frozen analysis supports the prespecified practical-gain claim. This statement is deliberately stronger than asking whether CFR-WM's point estimate is favorable: H1 required the lower bootstrap endpoint to clear a 5% practical margin and the exact seed-level test to reject that same margin. The distinction matters with eight top-level realizations, for which apparently stable trajectory-level averages can coexist with substantial training-seed uncertainty. The result therefore answers a narrow question about equal-trust context correction on the fixed benchmark; it is neither a universal ranking of world-model architectures nor a post hoc selection of the most favorable comparator.

The regime decomposition gives the proposed mechanism a more specific stress test. The corrupt/shifted estimate was 12.05%, whereas the clean estimate was -13.08%; after hierarchical gating, the full analysis does not support the gated robustness claim. Context corruption and parameter shift stress different terms in the error recursion. Noise and missingness weaken the measurement term in Equation (12), while parameter shift biases the nominal pre-update error and the subsequent nominal rollout. A gain of 0.5 cannot adapt between these cases. Consequently, similar aggregate effects can mask different cell patterns. The three cells with more than 10% degradation were

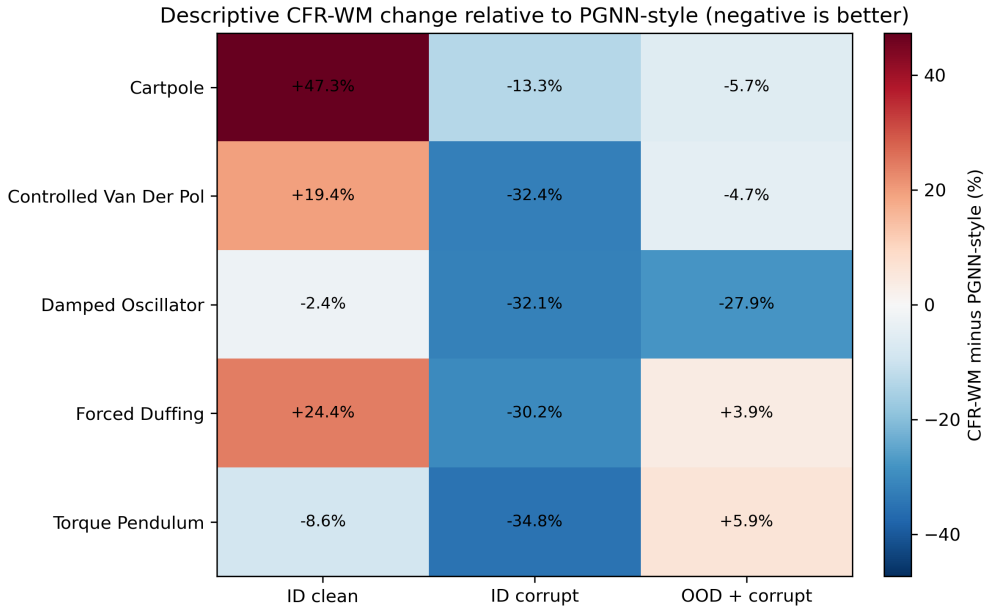


Fig. 3 Descriptive CFR-WM change relative to PGNN-style in each fixed benchmark cell. Negative values mean lower NRMSE-25 for CFR-WM.

all clean, led by cart–pole, while every ID-corrupt cell improved descriptively. This pattern is consistent with noise suppression within the bundled corrupt regimes, but the design cannot separate Gaussian noise from missingness or parameter shift; equal trust can also over-correct already accurate clean context. This is why the regime effects and all 15 cells are retained without promoting them to independent confirmatory or causal claims.

The gain sweep further prevents the fixed value from being interpreted as an optimal filter. The descriptively lowest aggregate NRMSE among the five frozen gains occurred at 0.50. This result locates equal trust within a continuum from pure nominal context propagation (gain 0) to direct measurement replacement (gain 1). It does not estimate a Bayes-optimal gain, because the noise, nominal-model error, and missingness processes vary by channel and trajectory. A learned gain was considered during development, but its advantage over the matched fixed-gain model missed a prespecified mechanism threshold; excluding it from confirmation avoids relabeling residual-model capacity as filter adaptation.

Multi-step training addresses a different part of the recursion. CFR-WM's relative effect against the one-step-trained variant was 59.38% (95% CI [48.88%, 68.88%]). The context filter changes the initial-error term, whereas multi-step likelihood exposes the residual transition to its own forecast distribution and targets accumulated local errors. Their roles are therefore complementary in the mathematical sense of the bound, even if the Holm-adjusted sensitivity contrast does not support a population-level component claim.

Relative to recent world-model and physics-guided work, the value of CFR-WM is auditability rather than architectural scale. It uses a transparent measurement rule, explicit missing-channel semantics, known-control free rollout, and a fixed practical-effect gate. This design trades adaptive expressivity for a mechanism that can be falsified with approximately parameter-matched gains. The archived failure of two earlier candidates is part of that evidence: it shows that physics, uncertainty, or a learned gate cannot be treated as useful merely because each is plausible in isolation. Future extensions should adopt the same separation when adding visual encoders, structured occlusion, ensembles or conformal calibration, and closed-loop planning [7, 41].

Most importantly, the study distinguishes mathematical, simulated, and deployment evidence. The filter identity is conditional; the benchmark establishes measured predictive effects under explicit perturbations; neither establishes causal mechanism, policy benefit, or hardware safety. Maintaining those boundaries is necessary for a compact physical-world-model result to be scientifically useful rather than rhetorically broader than its data.

7 Limitations

The evidential scope is intentionally narrower than the general world-model problem. First, all trajectories come from five analytically specified, low-dimensional controlled systems. This gives exact split, corruption, and parameter-shift provenance, but it does not reproduce the visual ambiguity, contacts, unmodeled objects, sensor synchronization failures, or nonstationary actuation encountered by embodied agents. The state is directly observed before synthetic corruption, future controls are supplied, and no policy is trained. Prediction quality here therefore cannot be converted into a claim about control return, planning safety, or real-robot transfer.

Second, every model receives the correct differential-equation family with imperfect parameters. Structural misspecification, hidden state variables, and informative sensor dropout may change the balance between nominal propagation and measurement correction. The gain 0.5 is deliberately fixed rather than estimated from uncertainty; the gain sweep measures sensitivity on this benchmark but does not establish filter optimality. The Gaussian diagonal forecast likewise cannot represent multimodality or full cross-state covariance, and the bounded one-step residual does not guarantee stable multi-step trajectories.

The two corrupt evaluation regimes bundle Gaussian observation noise and missingness, and the OOD-corrupt regime additionally bundles parameter shift. There are no noise-only, missingness-only, or clean-OOD factorial cells. Accordingly, regime differences cannot identify which perturbation caused an effect or establish the complementary update as its causal mechanism.

Third, inference has only eight independent seed-level training/data realizations. The exact sign-flip test is attainable for the prespecified five-comparison families, but it has coarse resolution and assumes symmetry of seed effects. Trajectory resampling improves precision for the fixed benchmark without creating additional independent

training replicates. System–regime contrasts are consequently descriptive, and all generalization claims remain limited to the frozen five-system grid. Larger preregistered seed studies, structurally wrong physics, state-dependent missingness, visual encoders, and hardware-in-the-loop evaluation are necessary before deployment claims would be justified.

8 Conclusion

We introduced CFR-WM, a compact world model that combines a prespecified, mask-exact complementary context update with a probabilistic physics-residual transition. The design isolates two sources of long-horizon error: uncertainty in the state at the forecast origin and error accumulated by the free-rollout transition. A conditional error identity explains when the fixed update can help, while gain and training-horizon sensitivities test its limits without claiming universal filtering optimality.

The confirmatory study trained 440 models across five controlled nonlinear systems, three deployment regimes, eight independent seed-level realizations, five broad comparators, and five sensitivities. Against the preselected PGNN-style baseline, the seed-balanced relative NRMSE-25 effect was 8.97% (95% CI [6.88%, 10.95%]); accordingly, H1 was supported, but the prespecified H2 robustness claim was not supported because clean noninferiority failed. Specifically, the corrupt/shifted component improved while clean-regime noninferiority failed, exposing the cost of a nonadaptive equal-trust update. These decisions follow the frozen 5% practical margin and exact seed-level tests, not a favorable pooled mean or selected system.

The work contributes both a technical composition and an evidential pattern: simple physics-aware corrections should be evaluated against matched sensitivities, and unsuccessful development mechanisms should remain visible rather than being absorbed into confirmation. The present evidence is limited to state-transition prediction with known controls on a synthetic-mechanistic benchmark. The next defensible steps are preregistered evaluation with structurally misspecified physics and informative missingness, followed by visual state estimation and hardware-in-the-loop closed-loop control. Those extensions require new safety, calibration, and policy endpoints; they cannot be inferred from NRMSE alone.

Declarations

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Competing interests The authors declare no competing interests.

Ethics approval and consent to participate Not applicable. The study uses analytically simulated controlled dynamics and no human participants, animals, or personal data.

Consent for publication Not applicable.

Data and code availability All trajectories are generated from the controlled differential equations supplied with the source package. The package contains generation seeds, split identifiers, checkpoints, trajectory-level metrics, the frozen confirmatory contract, and scripts for every table and figure. A public archival repository and persistent identifier must be added before submission.

Author contributions [To be completed by the authors before submission.]

Generative-AI assistance Generative-AI tools assisted drafting and software development. The authors retain responsibility for the final text and verified the equations, source metadata, code outputs, statistical results, and claims against the archived artifacts. This statement must be reconciled with the publisher policy in force on the submission date.

Appendix A Benchmark specification

Table A1 records the simulator time step and nominal parameterization; Table A2 gives every sampled training and OOD interval. All OOD intervals are non-overlapping on every named shifted parameter. The ranges are benchmark definitions, not a claim that they exhaust physically meaningful behavior.

Table A1 Controlled-system definitions. u is scalar in every system.

System	State	Δt	Nominal parameters
Damped oscillator	$(q, \dot{q})$	0.050	$m = 1.0,\ k = 1.6,\ c = 0.18$
Forced Duffing	$(q, \dot{q})$	0.040	$\alpha = -1.0,\ \beta = 1.0,\ c = 0.25$
Controlled Van der Pol	$(q, \dot{q})$	0.035	$\mu = 1.2,\ g_u = 1.0$
Torque pendulum	$(\sin\theta, \cos\theta, \dot{\theta})$	0.040	$g = 9.81,\ l = 1.0,\ m = 1.0,\ c = 0.12$
Cart–pole	$(x, \dot{x}, \theta, \dot{\theta})$	0.025	$g = 9.81,\ m_c = 1.0,\ m_p = 0.1,\ l/2 = 0.5,\ c = 0.04$

Table A2 Complete independently sampled physical-parameter intervals. Symbols follow Table A1; g_u is the Van der Pol control gain.

System	Training interval	OOD interval
Damped oscillator	$m \in [0.90, 1.10],\ k \in [1.40, 1.80],\ c \in [0.14, 0.22]$	$m \in [1.25, 1.45],\ k \in [2.05, 2.35],\ c \in [0.26, 0.34]$
Forced Duffing	$\alpha \in [-1.10, -0.90],\ \beta \in [0.90, 1.10],\ c \in [0.20, 0.30]$	$\alpha \in [-0.75, -0.60],\ \beta \in [1.25, 1.45],\ c \in [0.34, 0.44]$
Controlled Van der Pol	$\mu \in [1.00, 1.40],\ g_u \in [0.90, 1.10]$	$\mu \in [1.65, 1.95],\ g_u \in [1.25, 1.45]$
Torque pendulum	$g \in [9.65, 9.95],\ l \in [0.90, 1.10],\ m \in [0.90, 1.10],\ c \in [0.08, 0.16]$	$g \in [10.25, 10.55],\ l \in [1.25, 1.45],\ m \in [1.25, 1.45],\ c \in [0.20, 0.28]$
Cart–pole	$g \in [9.65, 9.95],\ m_c \in [0.90, 1.10],\ m_p \in [0.085, 0.115],\ l/2 \in [0.45, 0.55],\ c \in [0.02, 0.06]$	$g \in [10.25, 10.55],\ m_c \in [1.25, 1.45],\ m_p \in [0.135, 0.160],\ l/2 \in [0.65, 0.78],\ c \in [0.075, 0.105]$

The action generator is a bounded first-order filtered process. At step t, a Gaussian innovation with scale 0.55 times the system action limit is combined with the previous action using coefficients 0.18 and 0.82, then clipped. Limits are 1.5, 1.8, 1.2, 2.2, and 8.0 for the systems in Table A1. This produces persistent excitation without using the simulated state as feedback. It is part of the input distribution, not a learned behavior policy.

Train/validation/test separation occurs before window construction. A root NumPy `SeedSequence` spawns distinct split generators, which in turn spawn one generator per trajectory. The OOD test has a separate root seed. Observation corruption uses another recorded seed after state generation. Consequently, changing an observation-noise level does not change the underlying clean ID trajectory, and all methods see exactly the same corruption draw within a condition.

Appendix B Development firewall and confirmatory grid

Two earlier candidates were screened before the present confirmation. The first, RGP-WM, was rejected after correcting an inappropriate trajectory-local normalization in the development analysis. The second, a learned-gain PFR-WM, was rejected by its frozen kill rule because its NRMSE-25 improvement over the prespecified gain-0.5 sensitivity was only 0.223%, below the required 2%. Neither candidate appears in the confirmatory grid. The fixed gain of 0.5 used by CFR-WM was selected from that prespecified development sensitivity before any confirmatory seed was generated. These development outcomes motivate the design but provide no evidence for the confirmatory claims.

The immutable grid comprises 11 methods, eight seeds (41, 53, 67, 79, 83, 97, 109, and 113), five systems, and three test regimes. It therefore contains 440 fitted models, 1,320 aggregate evaluation cells, and 126,720 trajectory-level rows. Each seed–system dataset contains 256 training, 64 validation, and 96 test trajectories of 72 time steps. Models receive eight context steps and predict 25 steps open loop. Training observations use noise equal to 3% of the fixed training-state scale and a nominal 10% independent missing probability; the corrupt regimes use 8% observation noise and a nominal 25% probability. The first time point of each generated trajectory is forced observed in every channel, with independent masks sampled at all subsequent trajectory times; a later sliding-window context can begin with an imputed value. ID-clean and ID-corrupt evaluations share trajectory identifiers; the OOD trajectory set is disjoint.

The five broad comparators are compact, protocol-matched RSSM-style, SimNorm-style, Transformer-style, PGNN-style, and PG-DMM-style transitions. They are not exact reproductions of the cited end-to-end systems. Five prespecified CFR sensitivities test gains 0, 0.25, 0.75, and 1, plus a one-step-trained gain-0.5 model. CFR-WM retains the unused learned-gain-head parameters so that fixed-gain comparisons do not obtain a parameter-count advantage by removing that head.

All methods ingest normalized context values, masks, log-one-plus elapsed times, and actions. Except for the Transformer-style encoder, context is encoded by a GRU; open-loop recurrent updates use a GRUCell. Every MLP has two hidden linear layers with SiLU activation. Table B3 lists the remaining architecture-specific choices.

Table B3 Exact compact transition architectures used in the confirmatory grid. The bounded correction is $0.25\tanh(\cdot)$ and all forecast covariances are diagonal.

Method	Width	Context representation	Forecast transition and variance
RSSM-style	80	GRU; sigmoid feature scaling	data-only MLP mean/log-variance; GRUCell rollout
SimNorm-style	80	GRU; eight-group simplex projection	data-only MLP mean/log-variance; GRUCell rollout
Transformer-style	48	two causal layers, four heads, FF width 96, GELU, dropout 0	data-only MLP mean/log-variance; GRUCell rollout
PGNN-style	84	GRU	nominal one-substep RK4 plus deterministic residual MLP; learned constant log-variance
PG-DMM-style	84	GRU	nominal one-substep RK4 plus residual mean/log-variance MLP
CFR-WM and gain sensitivities	72	GRU states plus mask-exact fixed-gain context update	nominal one-substep RK4 plus residual mean/log-variance MLP

For CFR-WM, observed context channels use gain 0.5 and missing channels use gain 0 exactly. The learned gain head remains instantiated but is inactive. The four gain sensitivities change only the observed-channel gain; the one-step sensitivity changes only the likelihood horizon.

Appendix C Frozen hyperparameter search

The same six learning-rate/weight-decay trials were run on development seed 11 for every system–baseline pair (Table C4). The stored reliability-loss weight is shown because it is part of the generic training configuration, but it is inactive for all confirmatory methods in this paper. Each baseline uses the validation-NLL winner in Table C5; CFR-WM and all CFR sensitivities inherit the PGNN-style winner for the same system.

Table C4 Six-trial frozen optimization grid.

Trial	Learning rate	Weight decay	Stored λ_{rel}
0	3×10^{-4}	10^{-6}	0.1
1	6×10^{-4}	10^{-5}	0.2
2	10^{-3}	10^{-5}	0.2
3	1.5×10^{-3}	10^{-4}	0.3
4	2×10^{-3}	10^{-5}	0.4
5	3×10^{-3}	10^{-4}	0.2

Table C5 Frozen winning trial index by system and protocol-matched baseline.

System	RSSM-style	SimNorm-style	Transformer-style	PGNN-style	PG-DMM-style
Damped oscillator	3	2	4	5	1
Forced Duffing	5	4	5	5	5
Controlled Van der Pol	5	5	5	5	0
Torque pendulum	5	5	5	5	4
Cart–pole	5	5	5	5	5

Appendix D Confirmatory estimands and decisions

H1 is the sole confirmation gate. It asks whether CFR-WM improves fixed-training-scale NRMSE-25 over the preselected PGNN-style comparator by at least 5%. Within each seed, PGNN-style and CFR-WM are averaged with equal weight over all 15 fixed system–regime cells; the primary point estimate is the arithmetic mean of the eight seed-specific relative improvements. H1 requires finite metrics, a 95% bootstrap lower endpoint of at least 5%, and a one-sided exact sign-flip $p < 0.05$ for the 5% margin.

H2 is a hierarchically gated robustness claim. Conditional on H1, the combined corrupt and parameter-shifted regimes must satisfy the same 5% improvement rule, while ID-clean performance must be noninferior under a signed relative margin of -5%. Both components require their corresponding bootstrap lower endpoint to meet or exceed the margin and a one-sided exact sign-flip $p < 0.05$. If H1 fails, H2 is not supported irrespective of its component estimates.

The bootstrap uses 5,000 replicates. Each replicate resamples the eight seed clusters globally with replacement and then resamples paired trajectory identifiers within

every retained seed–system block. A single sampled identifier vector is reused for ID-clean and ID-corrupt repeated measurements, whereas the disjoint OOD identifiers are sampled separately. All five systems and three regimes remain in every replicate, and baseline, candidate, and their relative ratio are recomputed within the replicate. Exact tests enumerate all 2^8 seed-level sign flips and therefore rely on a seed-effect symmetry assumption. The five broad-comparator contrasts and the five sensitivity contrasts form separate Holm-adjusted families. The 15 system–regime contrasts are descriptive and carry no cell-level significance claim.

Appendix E Integrity and scope boundaries

The frozen contract records the method list, seeds, data design, training configuration, success rules, and SHA-256 hashes of the training, evaluation, analysis, and tuning artifacts. The dedicated runner rejects command-line overrides and resumes only provenance-matched JSON/checkpoint pairs. The generic development runner rejects every frozen seed and the reserved confirmatory output directory. Analysis aborts unless the exact grid, finite metrics, trajectory pairing, disjoint OOD identifiers, 40 distinct seed–system data hashes, and stored provenance all validate.

The first frozen-analysis invocation failed before inferential resampling and wrote no result directory because the generic loader had not copied the existing top-level `parameter_count` metadata into the aggregate frame used for descriptive model-size reporting. The archived amendment was frozen before performance inspection. It adds only that field by a method–seed–system many-to-one join, verifies all 440 JSON/checkpoint pairs and the completed-run manifest, and then invokes the unchanged frozen inference functions. The failed log, original analysis hash, amendment code hash, and amendment-contract hash are included in the reproduction package. No trajectory metric, hypothesis, threshold, resampling seed, replicate count, multiplicity family, or decision changed.

The top-level independent units are the eight stochastic training/data realizations; trajectories are nested observations and are not treated as independent experimental replicates. Statistical claims therefore concern this prespecified finite benchmark and variation across the frozen seed realizations. They do not establish universal superiority across physical systems, hardware platforms, sensing stacks, control policies, or real-world deployment conditions.

Appendix F Complete descriptive cell contrasts

Table F6 reports every prespecified CFR-WM–PGNN-style system–regime contrast. Absolute intervals come from the paired nested bootstrap. Relative changes use PGNN-style as the denominator, with negative values favoring CFR-WM. These fixed-cell estimates are descriptive: no cell-level significance decision or multiplicity-adjusted p value was prespecified.

Table F6 All 15 descriptive CFR-WM versus PGNN-style NRMSE-25 contrasts.

System	Regime	CFR–PGNN-style [95% CI]	Relative change, %
Cartpole	ID clean	+0.0163 [+0.0103, +0.0225]	+47.33
Cartpole	ID corrupt	-0.0207 [-0.0281, -0.0136]	-13.33
Cartpole	OOD + corrupt	-0.0111 [-0.0194, -0.0027]	-5.66
Controlled Van Der Pol	ID clean	+0.0059 [-0.0022, +0.0149]	+19.36
Controlled Van Der Pol	ID corrupt	-0.0318 [-0.0381, -0.0250]	-32.39
Controlled Van Der Pol	OOD + corrupt	-0.0083 [-0.0148, -0.0016]	-4.74
Damped Oscillator	ID clean	-0.0011 [-0.0041, +0.0019]	-2.41
Damped Oscillator	ID corrupt	-0.0291 [-0.0341, -0.0244]	-32.12
Damped Oscillator	OOD + corrupt	-0.0257 [-0.0295, -0.0223]	-27.86
Forced Duffing	ID clean	+0.0094 [+0.0028, +0.0168]	+24.37
Forced Duffing	ID corrupt	-0.0295 [-0.0342, -0.0249]	-30.24
Forced Duffing	OOD + corrupt	+0.0094 [+0.0011, +0.0174]	+3.90
Torque Pendulum	ID clean	-0.0046 [-0.0082, -0.0005]	-8.59
Torque Pendulum	ID corrupt	-0.0378 [-0.0436, -0.0316]	-34.80
Torque Pendulum	OOD + corrupt	+0.0110 [+0.0055, +0.0165]	+5.94

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